AppDeveloperCon

EUROPE

Deploying Your Apps to Kubernetes Without the Boilerplate

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Deploying Your Apps to Kubernetes Without the Boilerplate

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Agenda

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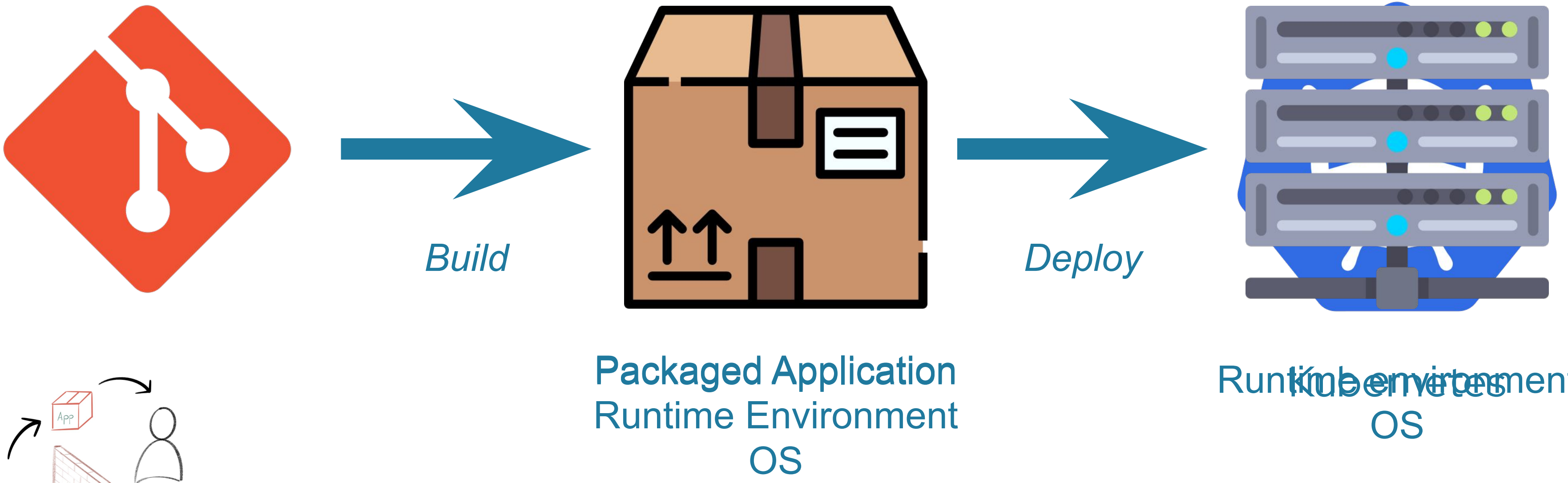
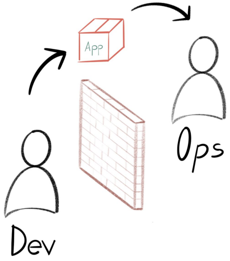
- Introduction (1 minute)
- Kubernetes: A Developer’s Dilemma (3 minutes)
  - Complexities of Kubernetes
  - Challenges developers face with Kubernetes Deployments
- Tooling Landscape: A Cross-Language Comparison (3 minutes)
- Simplifying Kubernetes Deployments: Tool Highlights (3 + 3 minutes)
- Deep Dive into Eclipse JKube (5 minutes)
- Live Demo: Deploying with Eclipse JKube (5 minutes)
- Q&A (2 minutes)

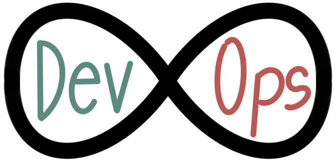
Agenda

- Problem statement:
  - Challenges application developers face when deploying to Kubernetes
  - *Shift left*
    - Container images - Dockerfiles
    - Cluster configuration manifests - YAMLs
    - Helm charts - more YAML
    - Security?
  - “Developers should focus on adding value to their product/project by solving customer/user requirements and problems. They shouldn’t focus or spend time on figuring out the platform”
-

Kubernetes: A Developer's Dilemma







Containerize

```
FROM openjdk:8-jdk-alpine
EXPOSE 8080
ARG JAR_FILE=target/*.jar
USER 1000:1000
COPY ${JAR_FILE} /deployments/app.jar
ENV JAVA_OPTIONS="-XX:MaxRAMPercentage=80"
ENTRYPOINT ["java","-jar","/deployments/app.jar"]
```

```
$ mvn package
$ docker build -t user/app:tag ./
$ docker login
$ docker push user/app:tag
```

YAMLs

```yaml
kind: Deployment
metadata:
  name: app
spec:
  replicas: 1
  selector:
# ...
```

```yaml
kind: Service
metadata:
  name: app
spec:
  ports:
# ...
```

```yaml
kind: ConfigMap
metadata:
  name: app
data:
 application.yml: >-
# ...
```

```yaml
kind: Secret
metadata:
  name: app
type: Opaque
data:
# ...
```

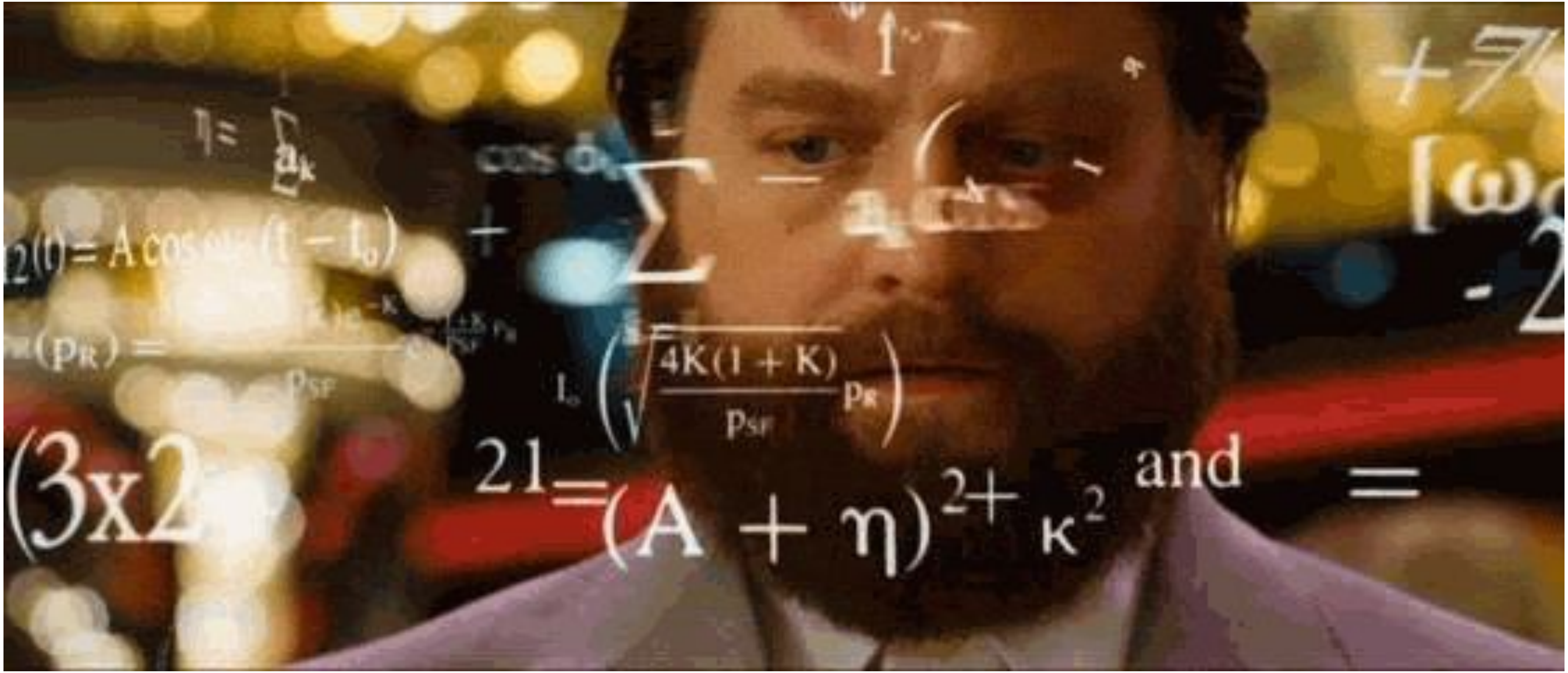
```
$ kubectl apply -f ./your-yaml.yml
```

From YAML FTW to WTF …


```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "7"
  creationTimestamp: "2024-03-12T14:05:01Z"
  generation: 7
  labels:
    control-plane: kserve-controller-manager
    controller-tools.k8s.io: "1.0"
  name: kserve-controller-manager
  namespace: kserve
  resourceVersion: "30553"
  uid: 473712e2-efa7-44dc-8c0e-f2e1de77ec07
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      control-plane: kserve-controller-manager
      controller-tools.k8s.io: "1.0"
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      annotations:
        kubectl.kubernetes.io/default-container: manager
      creationTimestamp: null
      labels:
        control-plane: kserve-controller-manager
        controller-tools.k8s.io: "1.0"
    spec:
      containers:
      - args:
        - --metrics-addr=127.0.0.1:8080
        - --leader-elect
        command:
        - /manager
        env:
        - name: POD_NAMESPACE
          valueFrom:
            fieldRef:
              apiVersion: v1
              fieldPath: metadata.namespace
        - name: SECRET_NAME
          value: kserve-webhook-server-cert
        image: kserve/kserve-controller:v0.12.0
        imagePullPolicy: Always
        livenessProbe:
          failureThreshold: 5
          httpGet:
            path: /healthz
            port: 8081
            scheme: HTTP
          initialDelaySeconds: 10
          periodSeconds: 10
          successThreshold: 1
          timeoutSeconds: 5
        name: manager
        ports:
        - containerPort: 9443
          name: webhook-server
          protocol: TCP
        readinessProbe:
          failureThreshold: 10
          httpGet:
            path: /readyz
            port: 8081
            scheme: HTTP
          initialDelaySeconds: 10
          periodSeconds: 5
          successThreshold: 1
          timeoutSeconds: 5
        resources:
          limits:
            cpu: 100m
            memory: 300Mi
          requests:
            cpu: 100m
            memory: 200Mi
        securityContext:
          allowPrivilegeEscalation: false
        terminationMessagePath: /dev/termination-log
        terminationMessagePolicy: File
        volumeMounts:
        - mountPath: /tmp/k8s-webhook-server/serving-certs
          name: cert
          readOnly: true
      - args:
        - --secure-listen-address=0.0.0.0:8443
        - --upstream=http://127.0.0.1:8080/
        - --logtostderr=true
        - --v=10
        image: gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1
        imagePullPolicy: IfNotPresent
        name: kube-rbac-proxy
        ports:
        - containerPort: 8443
          name: https
          protocol: TCP
        resources: {}
        terminationMessagePath: /dev/termination-log
        terminationMessagePolicy: File
      dnsPolicy: ClusterFirst
      restartPolicy: Always
      schedulerName: default-scheduler
      securityContext: {}
      serviceAccount: kserve-controller-manager
      serviceAccountName: kserve-controller-manager
      terminationGracePeriodSeconds: 10
      volumes:
      - name: cert
        secret:
          defaultMode: 420
          secretName: kserve-webhook-server-cert
```

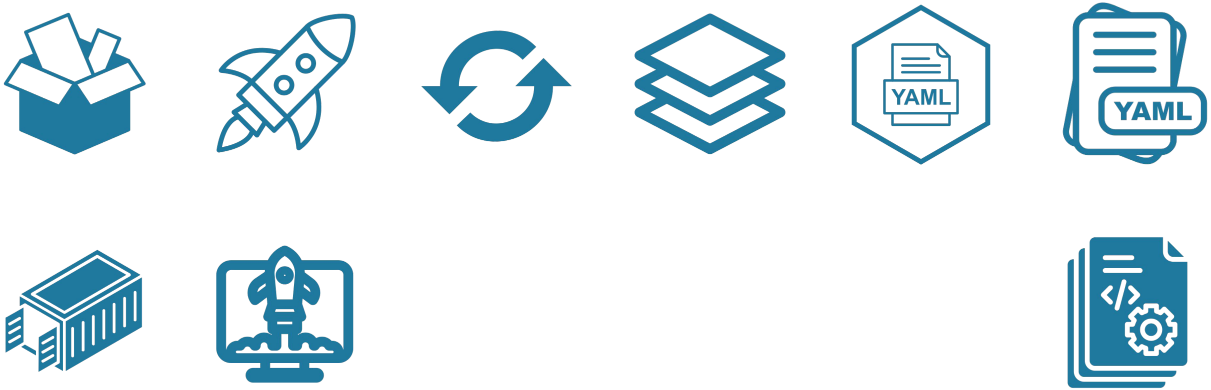

22:00

Kubernetes: A Developer’s Dilemma



Kubernetes Developer Tooling

- Huge landscape of Kubernetes Developer tooling for all kind of tasks
- Various degrees of abstraction
- Categorisation:
  - Cross-Language tools improving the developer workflow
  - Language-specific tools
- Main Tasks:
  - Building / packaging an application as OCI images
  - Deploying to Kubernetes
- Inner vs. Outer Loop
  - Inner loop during development (in development environment)
  - Outer loop preparing for deployment (in staging/production environment)
- Kubernetes resource manifests can become overwhelming
  - “YAML Stealth Level” : How much a tool can abstract-away Kubernetes YAML

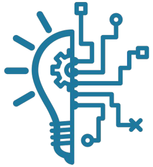

YAML
YAML

Kubernetes Developer Tooling

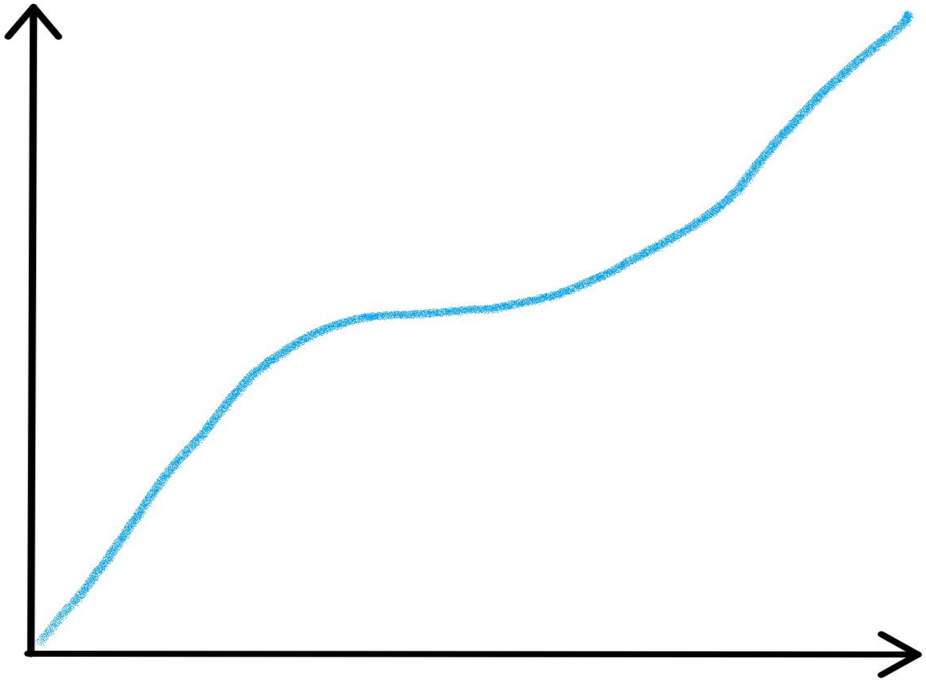
- Developer-oriented abstractions
- Building container images
- Deploying to Kubernetes
- Inner and Outer Loop
- YAML Stealth Level

Yaml Stealth Level - YSL

Simple
Opinionated
Restricted
Opaque Access



Complex
Neutral
Flexible
Transparent Access



Yaml Stealth Level

Cross-Language Developer Tools

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	Backing Company	Build	Deploy	Maturity	Inner Loop	Outer Loop	YSL
Helm			✔	❋❋❋	❋	❋❋❋	❋
Garden	Garden Inc.	✔	✔	❋❋	❋❋	❋❋	❋❋
Okteto	Okteto Inc.	✔	✔	❋❋	❋❋❋	❋	❋❋
Tilt	Docker	✔	✔	❋❋	❋❋❋	❋❋	❋
Dagger	Dagger	✔	✔	❋❋	❋❋	❋❋❋	❋❋❋
Skaffold	Google	✔	✔	❋❋❋	❋❋❋	❋❋	❋
Knative Functions		✔	✔	❋❋	❋❋	❋	❋❋❋

Language Specific Developer Tooling

	Language	Backing Company	Build	Deploy	Maturity	Inner Loop	Outer Loop	YSL
ko	Go	Google	✔	✔	⎈⎈⎈	⎈⎈	⎈⎈	⎈⎈
Jib	Java	Google	✔		⎈⎈⎈	⎈	⎈⎈	—
JKube	Java	Eclipse	✔	✔	⎈⎈⎈	⎈⎈⎈	⎈⎈	⎈⎈⎈

JKube

Tools and Plugins

Generate container images

Generate and deploy configuration manifests

More

Deploy Java on K8s

Components

JKube Kit
Kubernetes Maven/Gradle Plugin
OpenShift Maven/Gradle Plugin

Easy integration into any Java project

```xml
<plugins>
 <plugin>
   <groupId>org.eclipse.jkube</groupId>
   <artifactId>kubernetes-maven-plugin</artifactId>
 </plugin>
</plugins>
```

```
$ mvn package k8s:build k8s:resource k8s:apply
```

```groovy
plugins {
 id 'org.eclipse.jkube.kubernetes' version '1.16.1'
}
```

```
$ gradle build k8sBuild k8sResource k8sApply
```

Maven

Gradle

Eclipse JKube advantages

- No additional tools needed
- Multiple build strategies
- Zero configuration
- Flexible configuration
- Support for multiple frameworks
- Designed for the inner and outer loop

Deep Dive into JKube

Inspect

Analyze

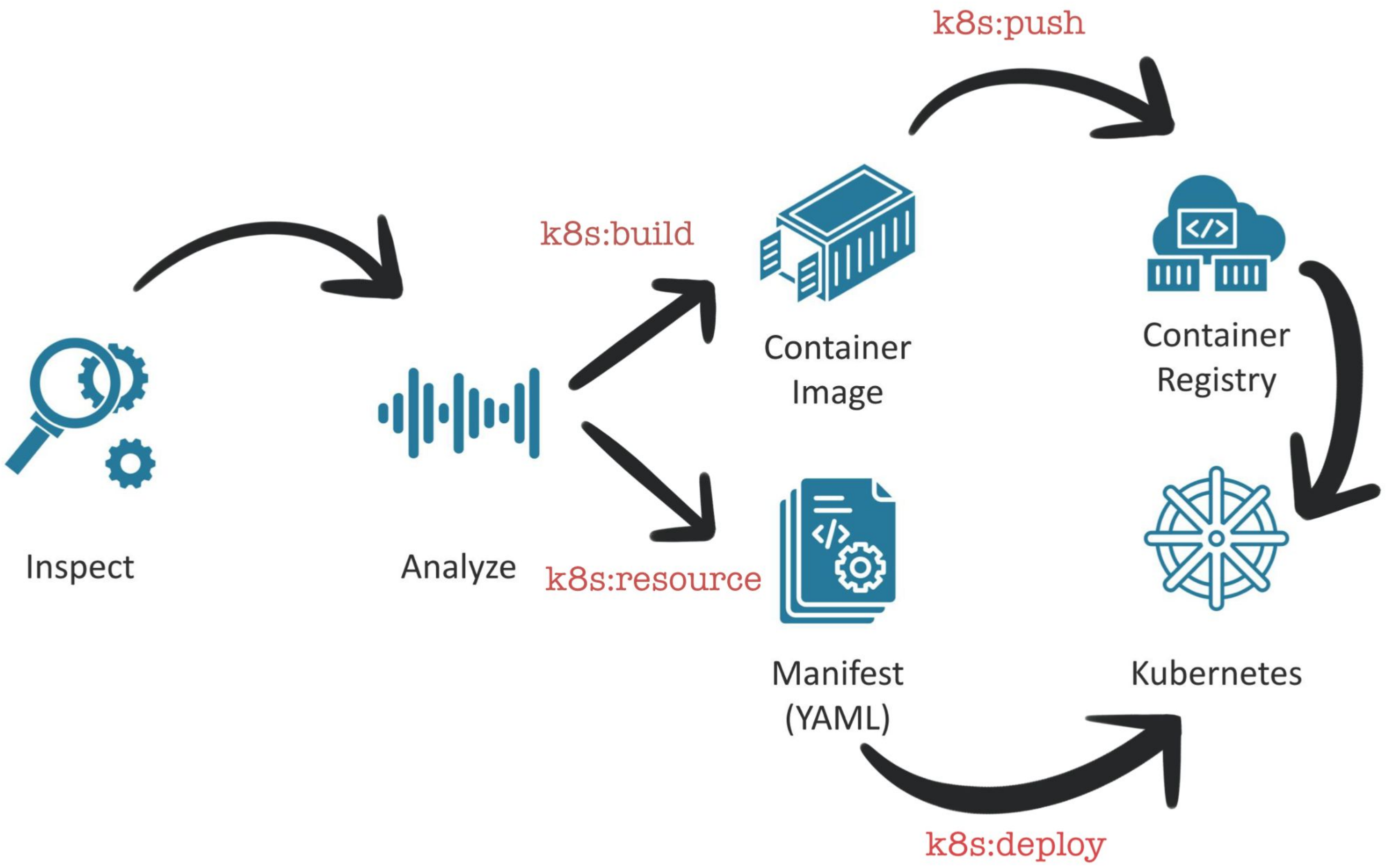
Container Image

Container Registry

Manifest (YAML)

Kubernetes

Deep Dive into JKube



Live Demo: Deploying with Eclipse JKube

Q&A